MILITARY STANDARD

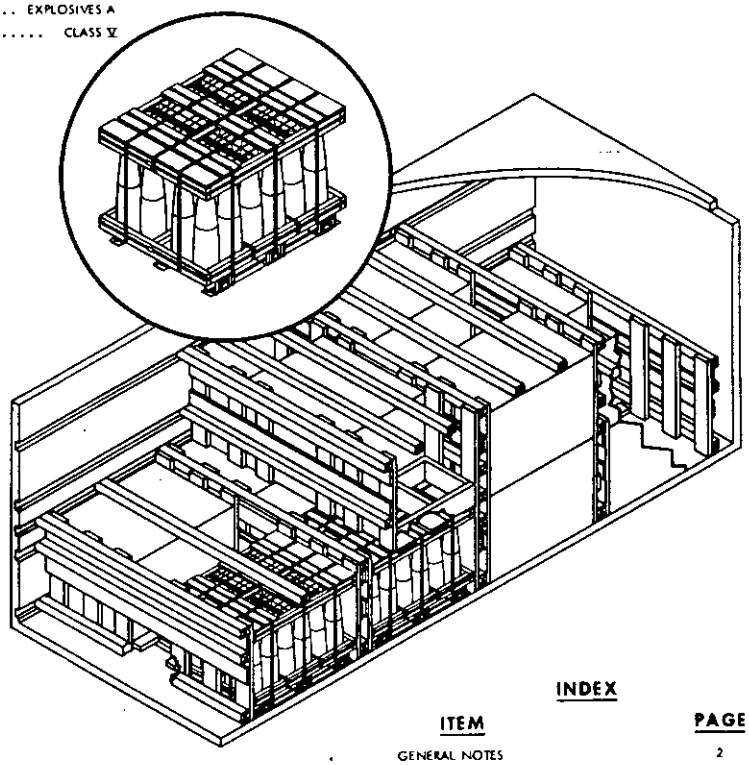
MIL-STD-1386-46
(NAVY)
5 NOVEMBER 1974

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

PROJECTILE,6" AAC, HC OR AP IN PALLET ADAPTER MK 4 AND MODS FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/111
DIMENSIONS - 6" AAC OR HC	46 3/8 L x 35 3/8 W x 35 5/8 H
6" AP	46 3/8 L x 35 3/8 W x 34 1/4 H
CUBE - 6" AAC OR HC	33.8 CU FT
6" AP	32.5 CU FT
WEIGHT (APPROX) - 6" AAC OR HC	2929 LBS
6" AP	3457 LBS
DOT HAZARD CLASSIFICATION	EXPLOSIVES A
COAST GUARD CLASS	CLASS V



INDEX

ITEM	PAGE
GENERAL NOTES	2
COFC & CONTAINERSHIP	2, 3, 4 & 5
DETAILS	6

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED FOR CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND MOVEMENT ABOARD DEPOT (NON-PUBLIC) ROADS ONLY.

M S Gray NWHL 11/5/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 11/5/74
SIGNATURE SEASYSCOM BY DIRECTION DATE

APPROVED BY US COAST GUARD

RJSchuring CAPT USCG
10-17-74

APPROVED BY BUREAU OF EXPLOSIVES

A. J. Grasmuck
10/24/74

ORIGINATOR L R Cairns
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 6" AAC, HC OR AP PROJECTILES IN PALLET ADAPTER MK 4 AND MODS. THE UNIT LOAD SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/111.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

CAUTION

WHEN POSITIONING CROSSMEMBERS MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO THE BELT RAILS FOR SHIPMENT.
5. SWAY BRACE, DETAIL N, PAGE 6, IS ASSEMBLED IN PLACE. PIECE 5, IS POSITIONED INBOARD OF THE OUTER LEGS OF THE PALLET SKIDS AND PIECE 6 IS POSITIONED AGAINST THE PALLET LEGS. NAIL PIECE 6 TO PIECE 5 WITH THREE 10d NAILS PER JOINT. CUT PIECE 7 TO FIT BETWEEN PIECE 6 AND NAIL PIECE 5 WITH THREE 10d NAILS.
6. WHEN THE LONGITUDINAL SLACK BETWEEN THE CROSSMEMBER AND THE SEPARATOR GATE IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND GATE, AS SHOWN IN DETAIL K, PAGE 6.
7. LENGTH OF STRUT, PIECE 9, SHALL BE SUFFICIENT TO CAUSE DETAIL C, TO FILL VOID BETWEEN THE UNIT LOAD AND THE CONTAINER SIDEWALL.
8. IF THE VOID BETWEEN THE TOP OF THE UNIT LOAD AND THE BOTTOM OF THE HOLD DOWN CROSSMEMBER EXCEEDS ONE HALF INCH, ADD FILL MATERIAL TO THE BOTTOM OF THE CROSSMEMBER AS SHOWN IN DETAIL J, PAGE 6. HOLD DOWN CROSSMEMBERS MUST BE POSITIONED OVER THE HARD POINTS OF UNIT LOAD.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER - FED. SPEC. - MM-L-751; NAILS FED. SPEC. - FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT.
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 VOL. 1.

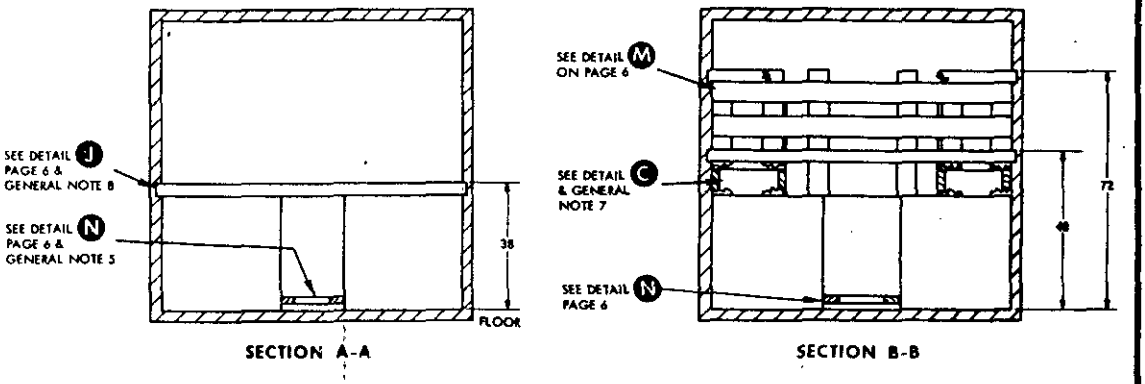
COFC AND CONTAINERSHIP - 6" AP PROJECTILES

1. THE CONTAINER LOAD CONSISTS OF 11 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED ACCORDING TO THIS DOCUMENT.

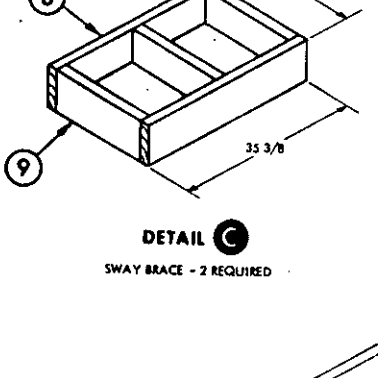
WARNING

BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MILVAN CHASSIS CANNOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

2. WHEN SHIPPING THE LOADED MILVAN CONTAINER COFC, THE FLAT CAR SHALL MEET THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. INSERT 1 x 6 x 36 3/8 SUPPORT PIECES, ITEM 10, UNDER UNIT LOAD STRAP AND STAPLE TO HOLD IN PLACE.

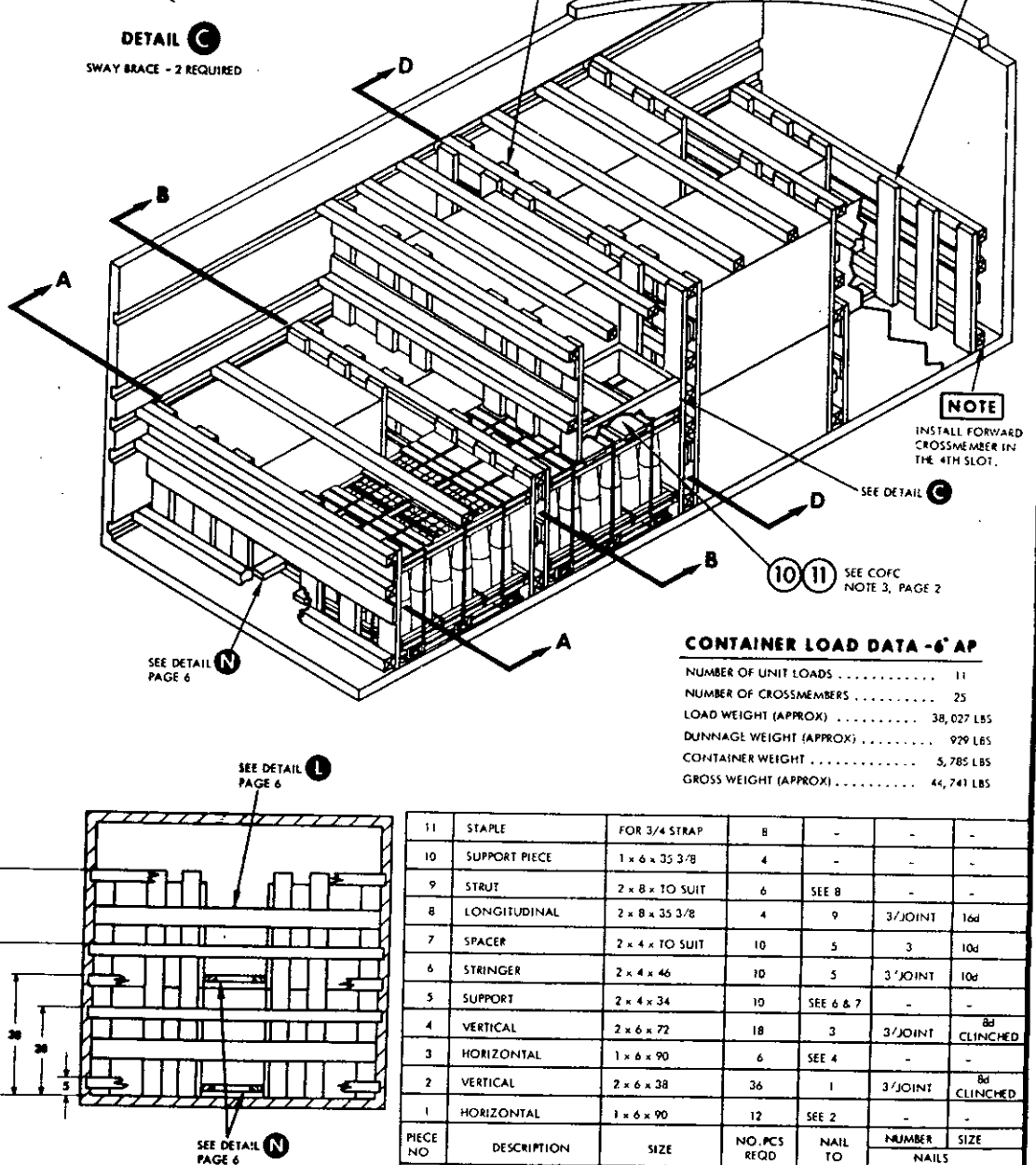


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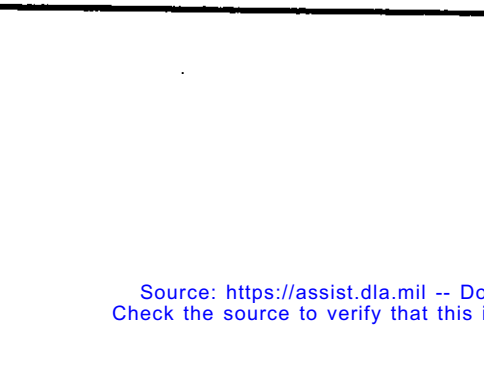
DETAIL C

SWAY BRACE - 2 REQUIRED



CONTAINER LOAD DATA -6" AP

NUMBER OF UNIT LOADS	11
NUMBER OF CROSSMEMBERS	25
LOAD WEIGHT (APPROX)	38,027 LBS
DUNNAGE WEIGHT (APPROX)	929 LBS
CONTAINER WEIGHT	5,785 LBS
GROSS WEIGHT (APPROX)	44,741 LBS



SECTION D-D

PIECE NO	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
11	STAPLE	FOR 3/4 STRAP	8	-	-	-
10	SUPPORT PIECE	1 x 6 x 35 3/8	4	-	-	-
9	STRUT	2 x 8 x TO SUIT	6	SEE 8	-	-
8	LONGITUDINAL	2 x 8 x 35 3/8	4	9	3/JOINT	16d
7	SPACER	2 x 4 x TO SUIT	10	5	3	10d
6	STRINGER	2 x 4 x 46	10	5	3/JOINT	10d
5	SUPPORT	2 x 4 x 34	10	SEE 6 & 7	-	-
4	VERTICAL	2 x 6 x 72	18	3	3/JOINT	8d CLINCHED
3	HORIZONTAL	1 x 6 x 90	6	SEE 4	-	-
2	VERTICAL	2 x 6 x 38	36	1	3/JOINT	8d CLINCHED
1	HORIZONTAL	1 x 6 x 90	12	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

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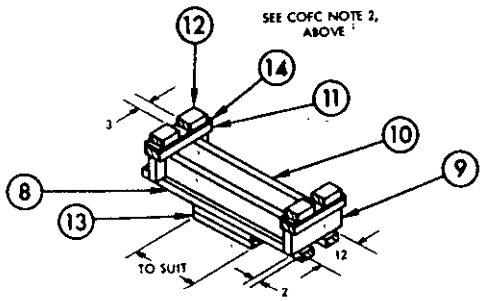
COFC AND CONTAINERSHIP -6" AAC OR HC PROJECTILES

1. THE CONTAINER LOAD CONSISTS OF 12 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED ACCORDING TO THIS DOCUMENT.

WARNING

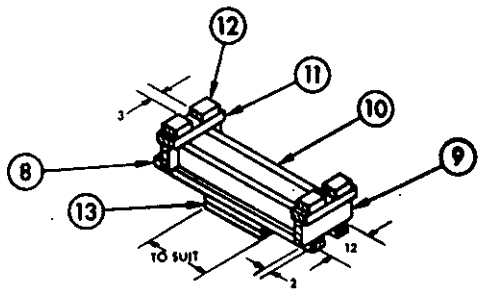
BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MILVAN CHASSIS CANNOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

2. HEIGHT OF HOLD DOWN ASSEMBLIES, DETAILS E AND F MAY BE VARIED TO ACCOMMODATE DIFFERENCES IN UNIT LOAD HEIGHTS BY ADJUSTING PIECES 11 AND 14 OR OMITTING PIECES AS REQUIRED. STRUCTURAL INTEGRITY OF HOLD DOWN ASSEMBLIES SHALL BE MAINTAINED. THE VOID BETWEEN HOLD DOWN CROSSMEMBER AND HOLD DOWN ASSEMBLY SHALL NOT EXCEED ONE HALF INCH.
3. WHEN LOAD BEARING MEMBER, PIECE 8 OF HOLD DOWN ASSEMBLY FALLS BETWEEN BATTENS ON TOP FRAME OF UNIT LOAD, INSERT 1 x 6 x 35 3/8 SUPPORT PIECES, ITEM 15, UNDER STRAPPING AND STAPLE TO HOLD IN PLACE.
4. WHEN SHIPPING THE LOADED MILVAN CONTAINER, COFC, THE FLAT CAR SHALL MEET THE REQUIREMENT OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.



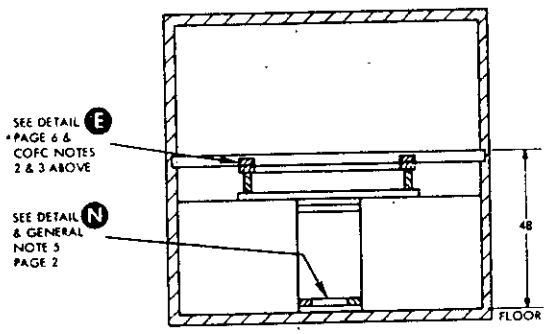
DETAIL E

HOLD DOWN ASSEMBLY
2 REQUIRED

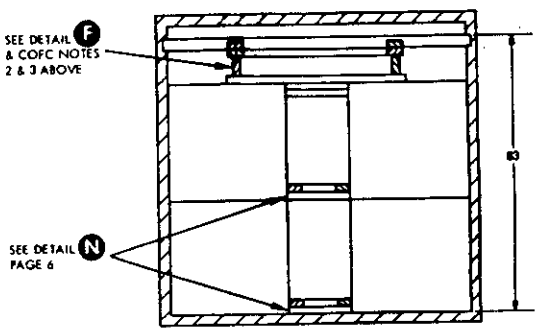


DETAIL F

HOLD DOWN ASSEMBLY
4 REQUIRED

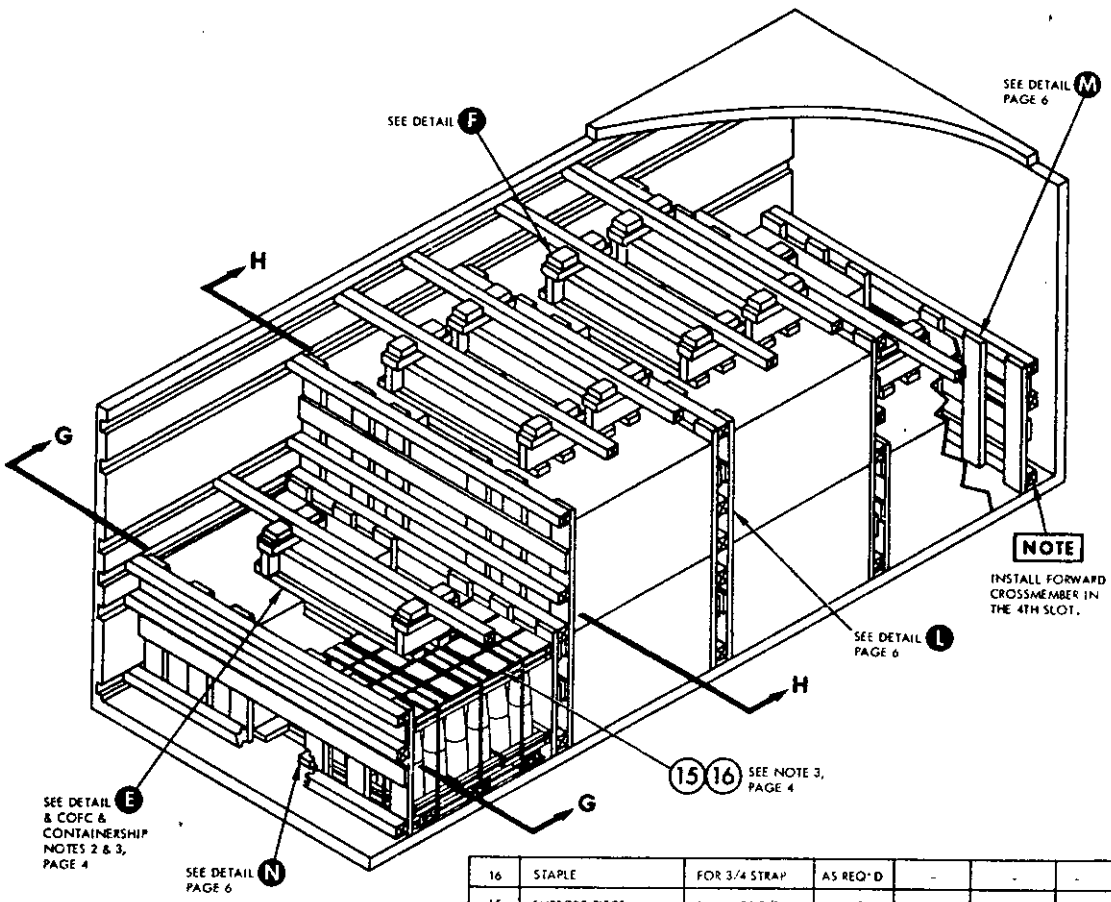


SECTION G-G



SECTION H-H

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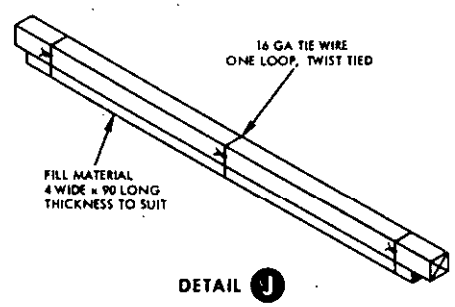
CONTAINER LOAD DATA-6˝ AAC OR HC

NUMBER OF UNIT LOADS	12
NUMBER OF CROSSMEMBERS	25
LOAD WEIGHT (APPROX)	35, 148 LBS
DUNNAGE WEIGHT (APPROX)	1, 176 LBS
CONTAINER WEIGHT	5, 785 LBS
GROSS WEIGHT (APPROX)	42, 109 LBS

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
16	STAPLE	FOR 3/4 STRAP	AS REQ'D	-	-	-
15	SUPPORT PIECE	1 x 6 x 35 3/8	AS REQ'D	-	-	-
14	SHIM	1 x 4 x 18	AS REQ'D	LAMINATE	-	-
13	SPACER	2 x 4 x TO SUIT	12	10	6	10d
12	RETAINER	2 x 4 x 6	24	13	3	10d
11	SHIM	2 x 4 x 18	12	11	3	10d
10	SUPPORT	2 x 6 x TO SUIT	12	SEE 10 & 11	-	-
9	RISER	2 x 6 x 17	12	12	-	-
8	LOW BEARING PIECE	2 x 4 x 54	12	12 11	3 2/JOINT	10d 10d
7	SPACER	2 x 4 x TO SUIT	12	5	3	10d
6	STRINGER	2 x 4 x 46	12	5	3/JOINT	10d
5	SUPPORT	2 x 4 x 34	12	SEE 6 & 7	-	-
4	VERTICAL	2 x 6 x 72	24	3	3/JOINT	8d CLINCHED
3	HORIZONTAL	1 x 6 x 90	8	SEE 4	-	-
2	VERTICAL	2 x 6 x 38	24	1	3/JOINT	8d CLINCHED
1	HORIZONTAL	1 x 6 x 90	8	SEE 2	-	-

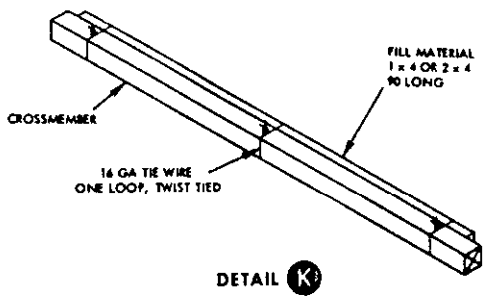
LIST OF MATERIALS AND NAILING DATA

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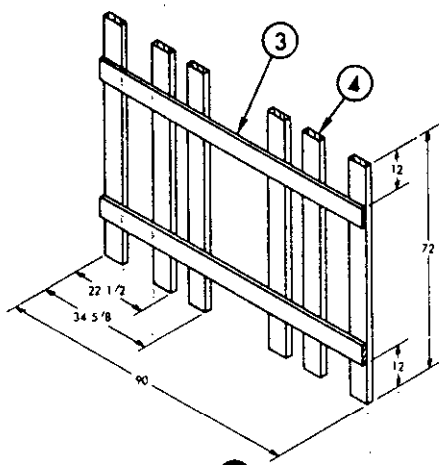


DETAIL J

(SEE GENERAL NOTE 8, PAGE 2)

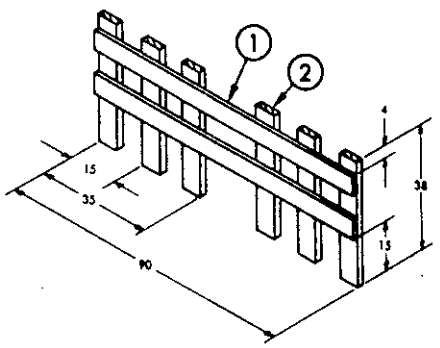


DETAIL K



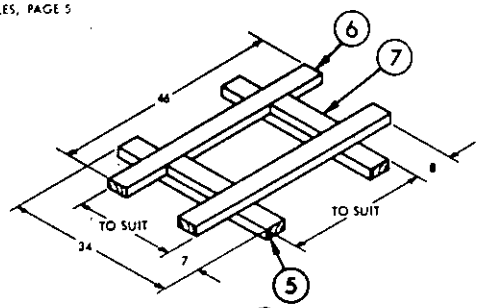
DETAIL L

SEPARATOR GATE
3 - REQUIRED F/AP PROJECTILES, PAGE 3
4 - REQUIRED F/AAC OR HC PROJECTILES, PAGE 5



DETAIL M

SEPARATOR GATE
6 - REQUIRED F/AP PROJECTILES, PAGE 3
4 - REQUIRED F/AAC OR HC PROJECTILES, PAGE 5



DETAIL N

SWAY BRACE
5 - REQUIRED F/AP PROJECTILES, PAGE 3
6 - REQUIRED F/AAC OR HC PROJECTILES, PAGE 5

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT NO. 8145-N-257

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